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from google.colab import files

uploaded = files.upload()

import pandas as pd

df = pd.read_csv('sentimentdataset.csv')

df.head()

plt.figure(figsize=(20,10))

sns.countplot(x='Sentiment', data=df)

plt.title("Distribution of Sentiment Labels")

plt.xticks(rotation=45)

plt.show()

X = df['Text']

y = df['Sentiment']

vectorizer = TfidfVectorizer(stop_words='english', max_features=5000)

X_vectorized = vectorizer.fit_transform(X)

X_train, X_test, y_train, y_test = train_test_split(X_vectorized, y, test_size=0.2, random_state=42)

model = LogisticRegression(max_iter=1000)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Classification Report:\n")

print(classification_report(y_test, y_pred))

plt.figure(figsize=(10,6))

cm = confusion_matrix(y_test, y_pred)

sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=model.classes_,
yticklabels=model.classes_)

plt.title("Confusion Matrix")

plt.xlabel("Predicted Sentiment")

plt.ylabel("Actual Sentiment")

plt.show()
```